



SpaceY into the Space Race

IBM Developer SKILLS NETWORK
Andrés Lizardo
July 2026

Full GitHub Project:

<https://github.com/lizardoom/Applied-Data-Science-Capstone.git>



Outline

- Summary
- Introduction
- Methodology
- Results
- Conclusion



Executive Summary

- Summary of methodologies
 - Data collection
 - Data wrangling
 - EDA with data visualization
 - EDA with SQL
 - Map with Folium
 - Dashboard with Plotly Dash
 - Predictive analysis
- Summary of all results
 - EDA results
 - Interactive analysis
 - Predictive analysis



Introduction

- Background and context
 - SpaceX advertises Falcon 9 rocket launches on its website, with a cost of 62 million dollars; other providers cost upward 165 million dollars each, much of the savings is because SpaceX can reuse the first stage
- Problems you want to solve
 - The project task is to predicting if the first stage of SpaceX Falcon 9 rocket will land successfully.

Methodology



Methodology

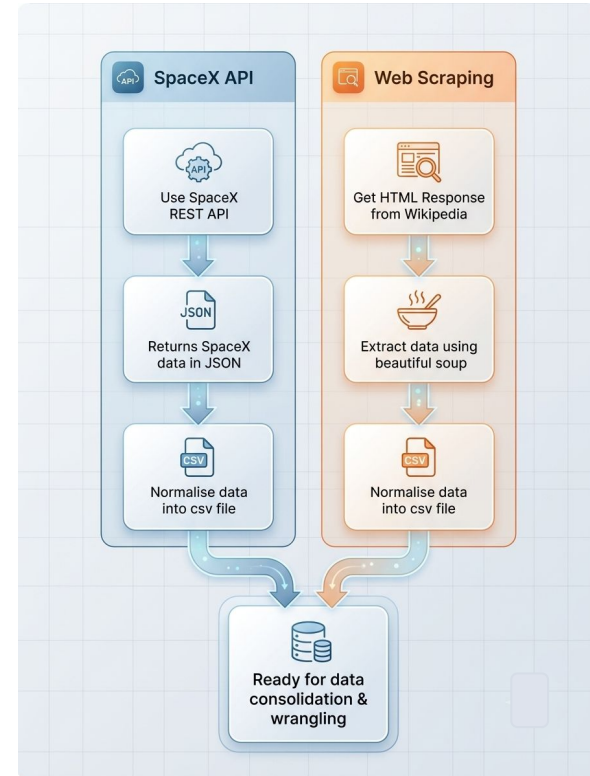
Executive Summary:

- Data collection methodology
 - SpaceX Rest API
 - Web Scraping from Wikipedia
- Data wrangling
- Exploratory data analysis using visualization and SQL
- Interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

- The following datasets was collected as it follows:
 - SpaceX launch data that is gathered from the SpaceX REST API
 - This API will give us data about launches, including information about the rocket used, payload, and other specs.
 - The other source for obtaining Falcon 9 Launch data is web scraping wikipedia using BeautifulSoup.

<https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - SpaceX API

Data collection with SpaceX REST calls.

On the image we have a workflow of the steps needed

<https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

1. Getting Response from API

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
response = requests.get(spacex_url).json()
```

2. Converting Response to a .json file

```
response = requests.get(static_json_url).json()
data = pd.json_normalize(response)
```

3. Apply custom functions to clean data

```
getLaunchSite(data)
getPayloadData(data)
getCoreData(data)
```

```
getBoosterVersion(data)
```

4. Assign list to dictionary then dataframe

```
launch_dict = {'FlightNumber': list(data['flight_number']),
               'Date': list(data['date']),
               'BoosterVersion': BoosterVersion,
               'PayloadMass': PayloadMass,
               'Orbit': Orbit,
               'LaunchSite': LaunchSite,
               'Outcome': Outcome,
               'Flights': Flights,
               'GridFins': GridFins,
               'Reused': Reused,
               'Legs': Legs,
               'LandingPad': LandingPad,
               'Block': Block,
               'ReusedCount': ReusedCount,
               'Serial': Serial,
               'Longitude': Longitude,
               'Latitude': Latitude}
```

```
df = pd.DataFrame.from_dict(launch_dict)
```

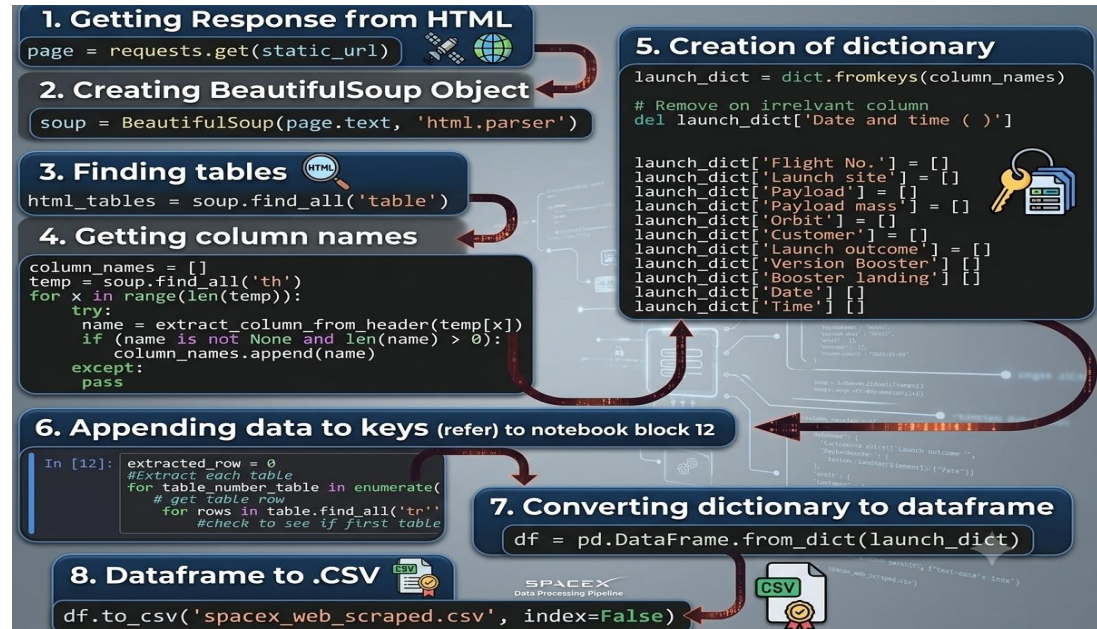
5. Filter dataframe and export to flat file (.csv)

```
data_falcon9 = df.loc[df['BoosterVersion']!="Falcon 1"]
data_falcon9.to_csv('dataset_part_1.csv', index=False)
```

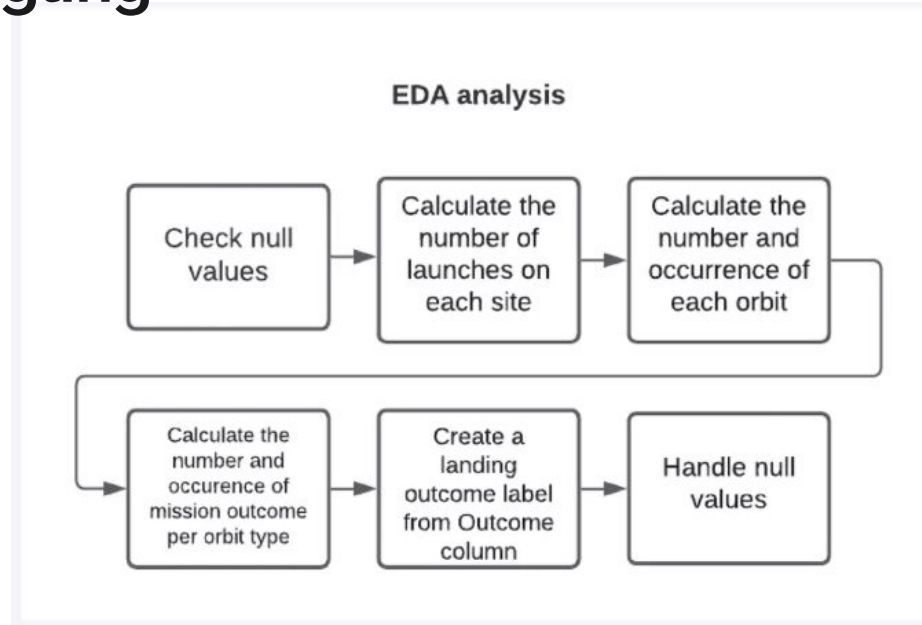

Data Collection - Scrapping

Web Scrapping from Wikipedia

<https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>

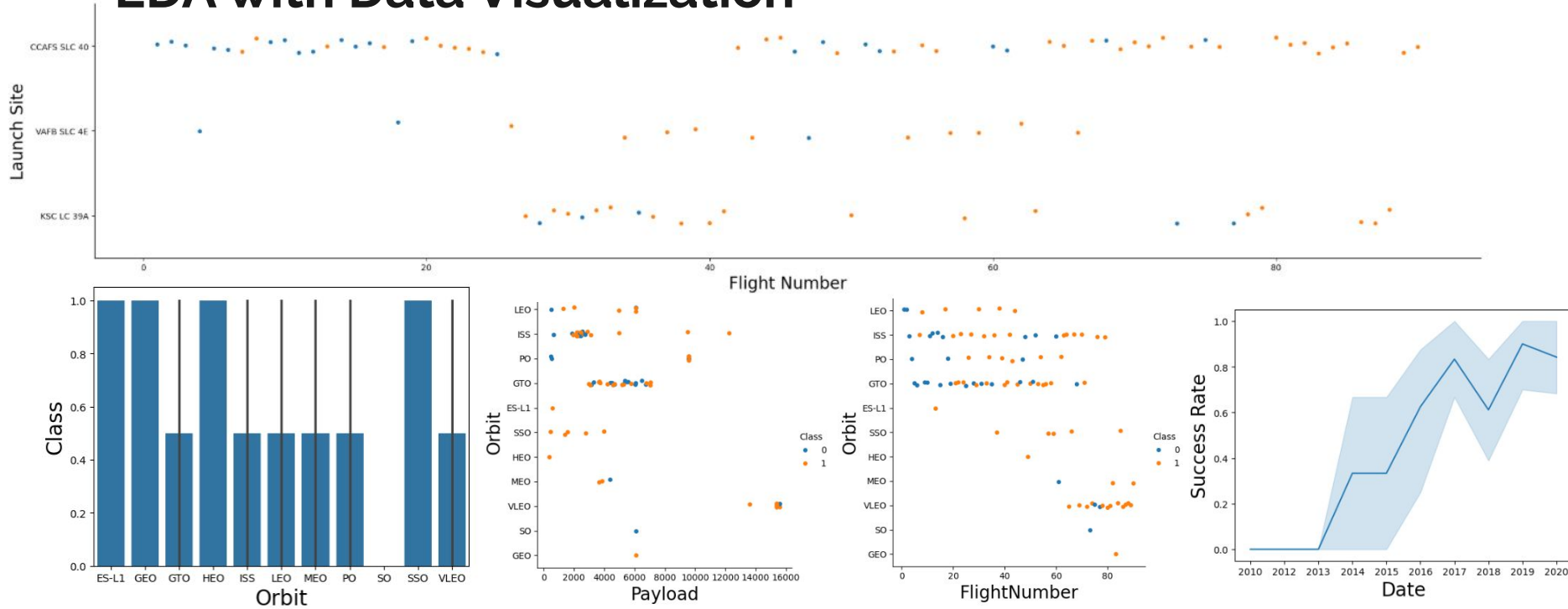


Data Wrangling



<https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/edadataviz.ipynb>

EDA with Data Visualization





EDA with SQL

- SQL queries performed include:
 - Displaying the names of the unique launch sites in the space mission
 - Displaying 5 records where launch sites begin with the string 'KSC'
 - Displaying the total payload mass carried by boosters launched by NASA (CRS)
 - Displaying average payload mass carried by booster version F9 v1.1
 - Listing the date where the successful landing outcome in drone ship was achieved.
 - Listing the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000
 - Listing the total number of successful and failure mission outcomes
 - Listing the names of the booster_versions which have carried the maximum payload mass.
 - Listing the records which will display the month names, successful landing_outcomes in ground pad ,booster versions, launch_site for the months in year 2017
 - Ranking the count of successful landing_outcomes between the date 2010 06 04 and 2017 03 20 in descendingorder.

Interactive Map with Foluim

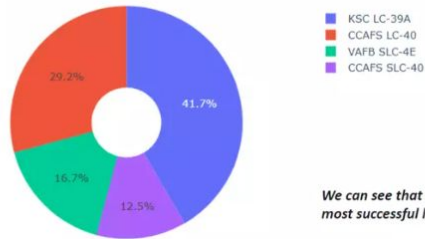


Map markers have been added to the map with aim to finding an optimal location for building a launch site

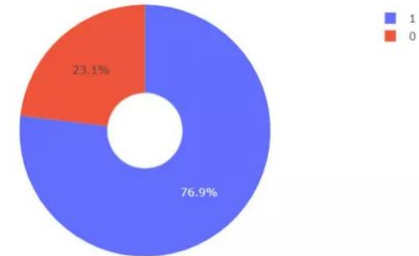
https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Dashboard with Plotly Dash

Total Success Launches By all sites

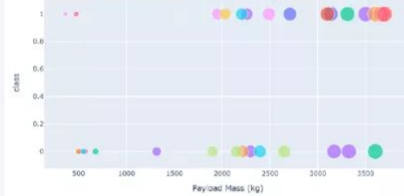


We can see that KSC LC-39A had the most successful launches from all the sites

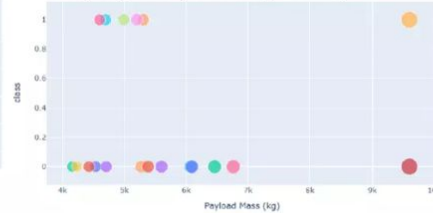


KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

Low Weighted Payload 0kg – 4000kg



Heavy Weighted Payload 4000kg – 10000kg

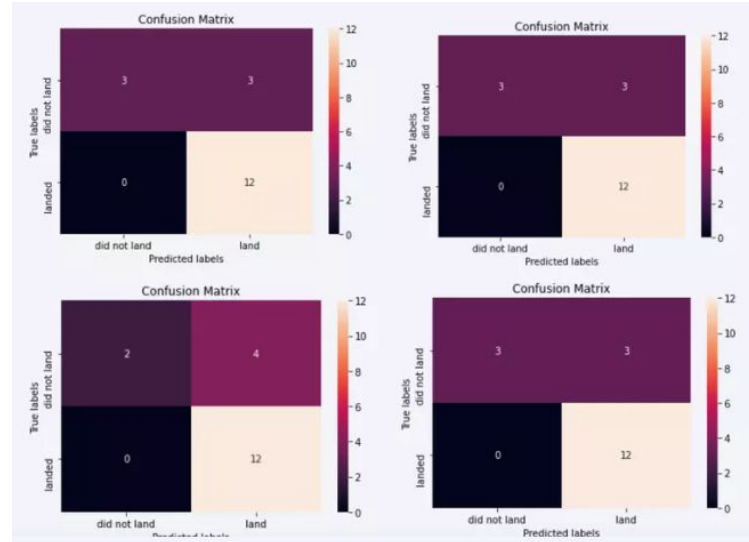
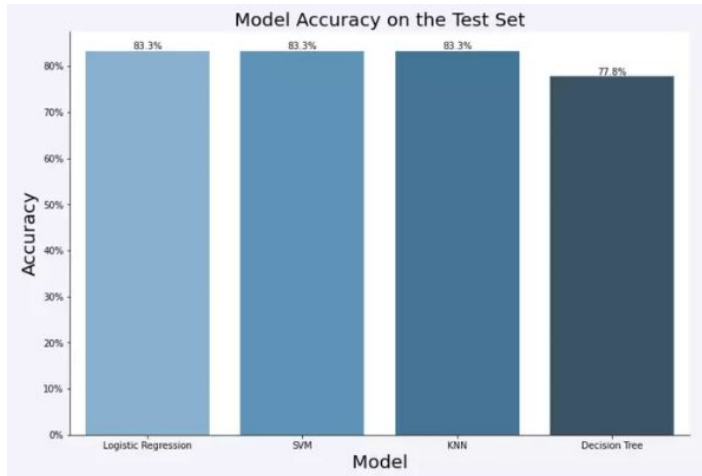


We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

[https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/SpaceX Machine%20Learning%20Prediction Part 5.ipynb](https://github.com/lizardoom/Applied-Data-Science-Capstone/blob/main/SpaceX%20Machine%20Learning%20Prediction%20Part%205.ipynb)

Predictive analysis

The SVM, KNN and Logistic Regression model achieved the highest accuracy at 83.3% while the SVM performs the best in terms of area under the curve at 0.958.





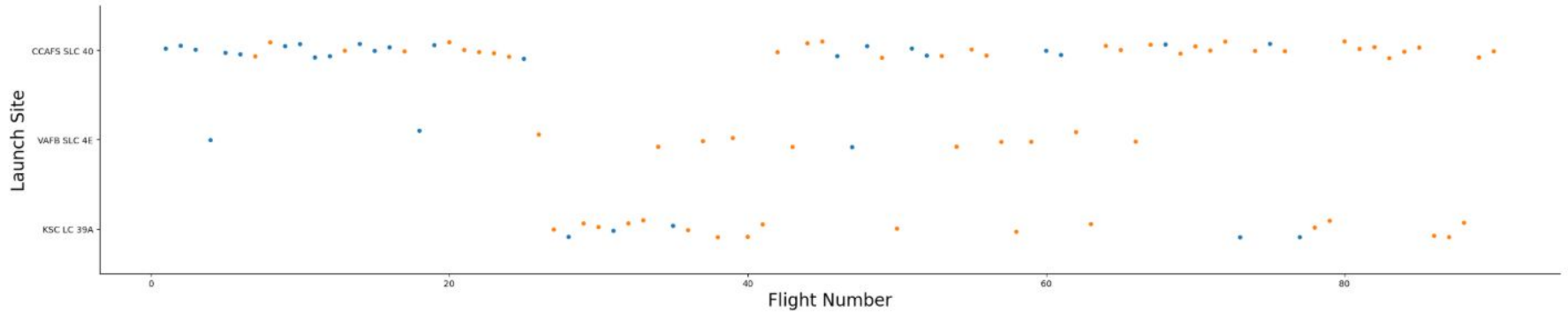
Results

- The SVM, KNN and Logistic Regression models are the best in terms of prediction, accuracy for this dataset
- Low weighted payloads perform better than the heavier payloads
- The success rates for SpaceX launches is directly proportional to the launch time, if this behaviour continues they will eventually perform perfect launches.
- KSC LC 39A had the most successful launches from all the sites
- Orbit GEO, HEO, SSO, ES L1 have the best success rate.

Insights drawn from EDA

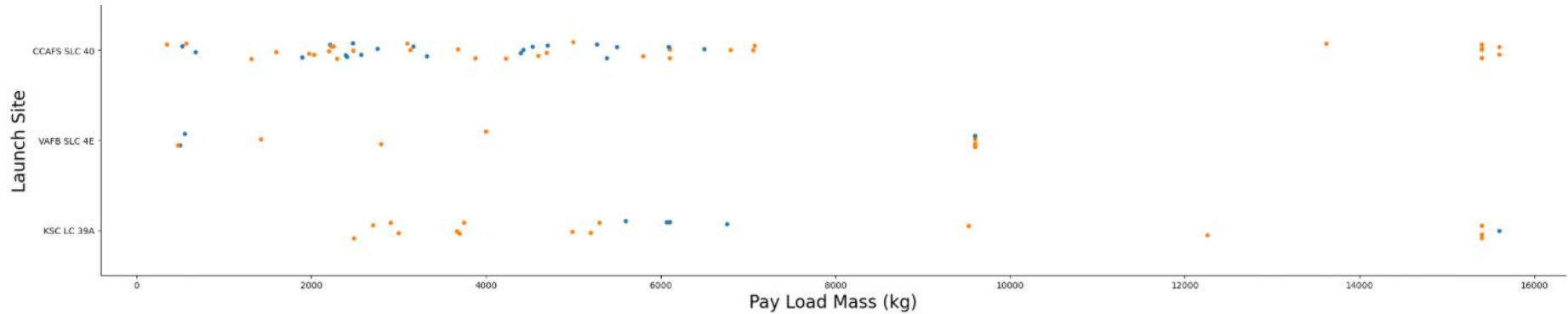


Flight Number vs Launch Site



Launches from CCAFS SLC 40 are significantly more than launches from the other two sites

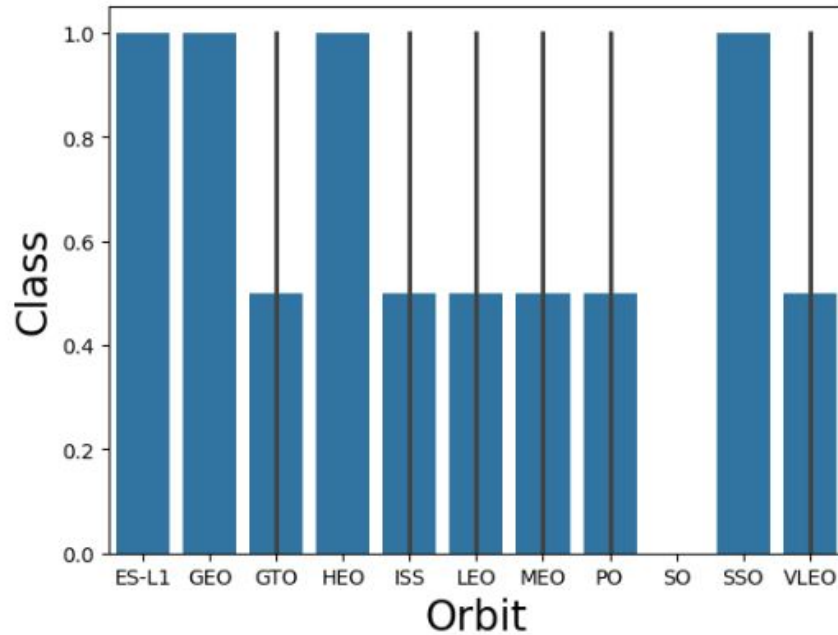
Payload vs Launch Site



The CCAFS SLS 40 owns the majority of the lower mass launches

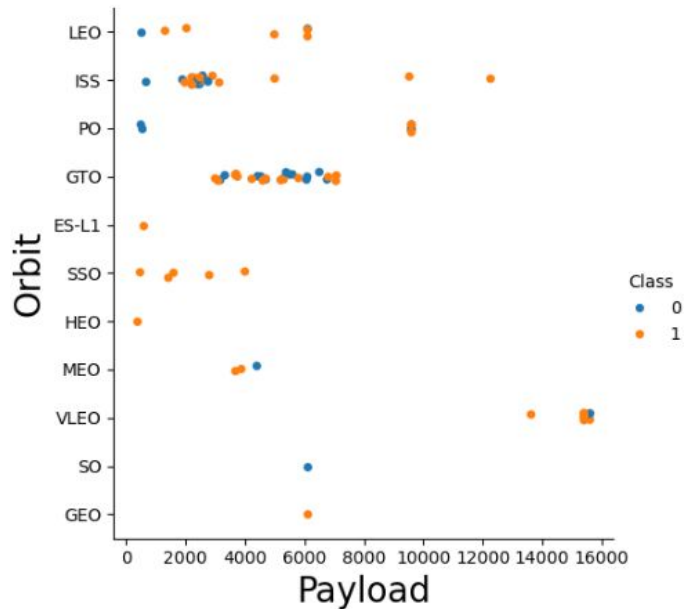


Success rate vs Orbit type



The orbit types of ES-L1, GEO, HEO, SSO, have the highest success rate

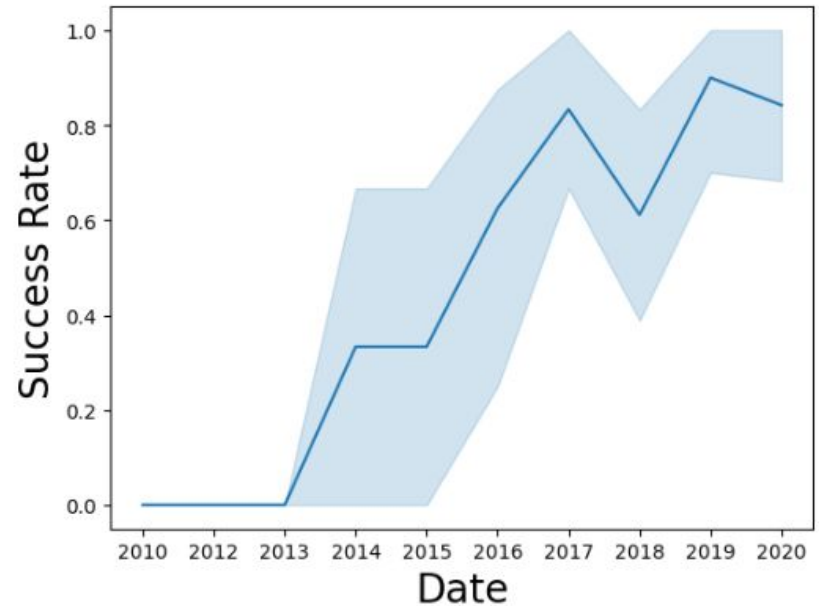
Payload vs Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

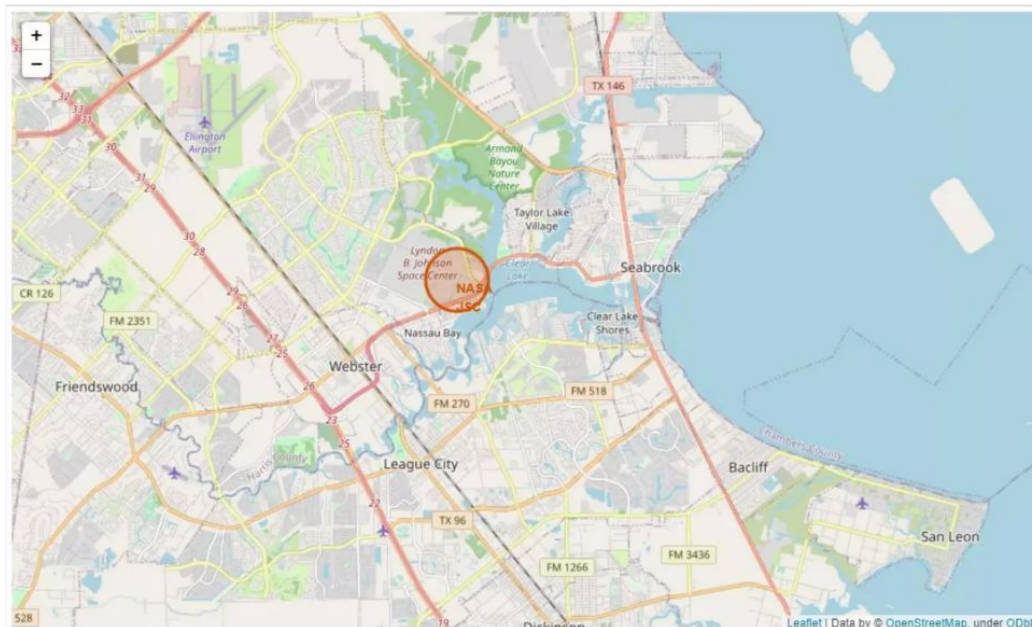
Launch success yearly trend

There is a significant growth in the success rate in the more recent years. There is a diminishing of success in 2018, which was immediately corrected to continue with the growth in the success.

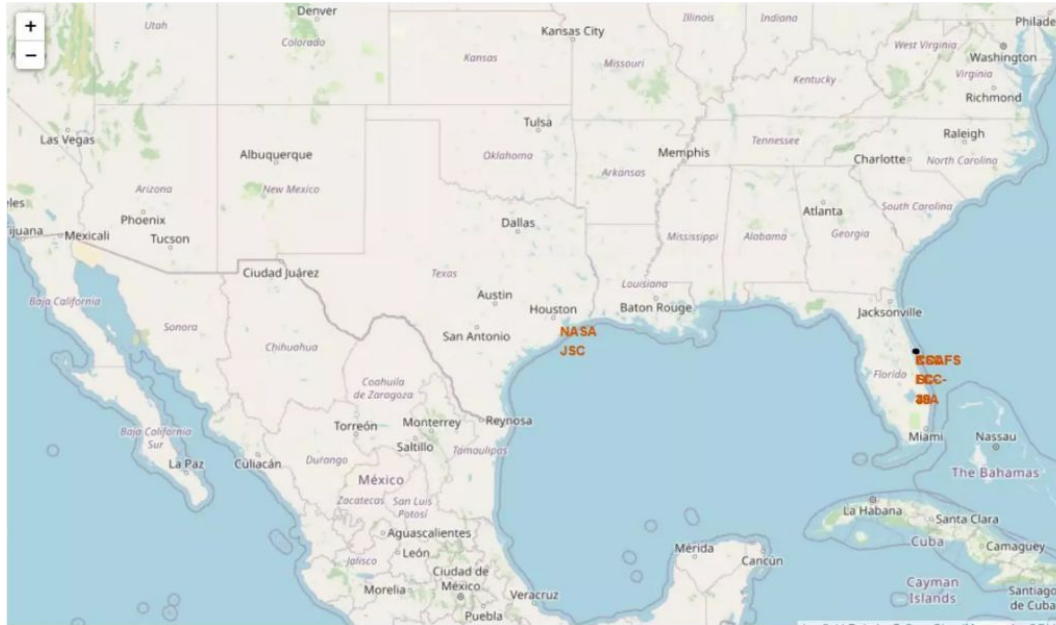


Launch Sites Proximities Analysis

Launch sites in the map



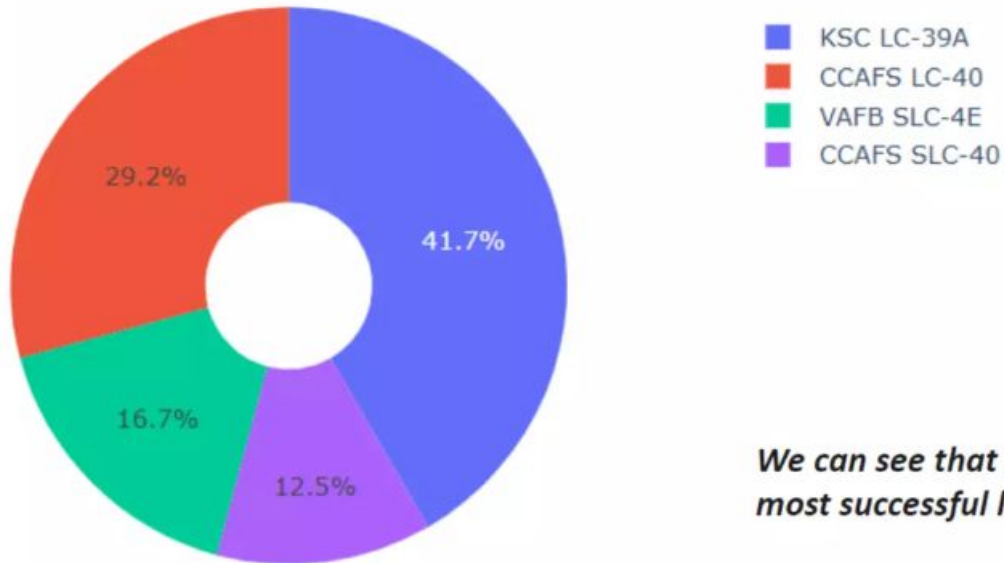
Success/Failed launches marked on the map



Distances between a launch site to its proximities

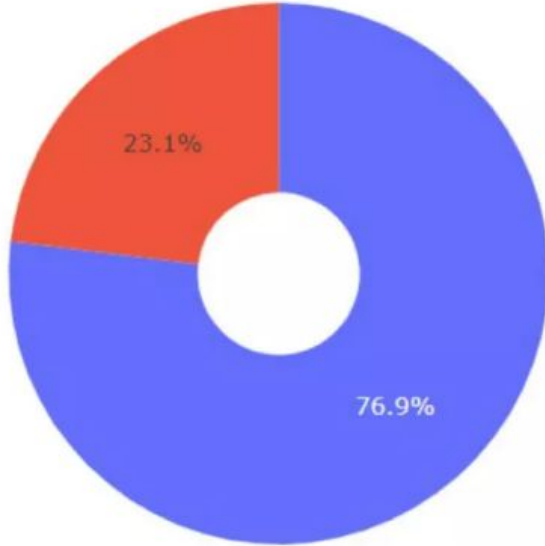
Dashboard with Plotly Dash

Total success launches by all sites



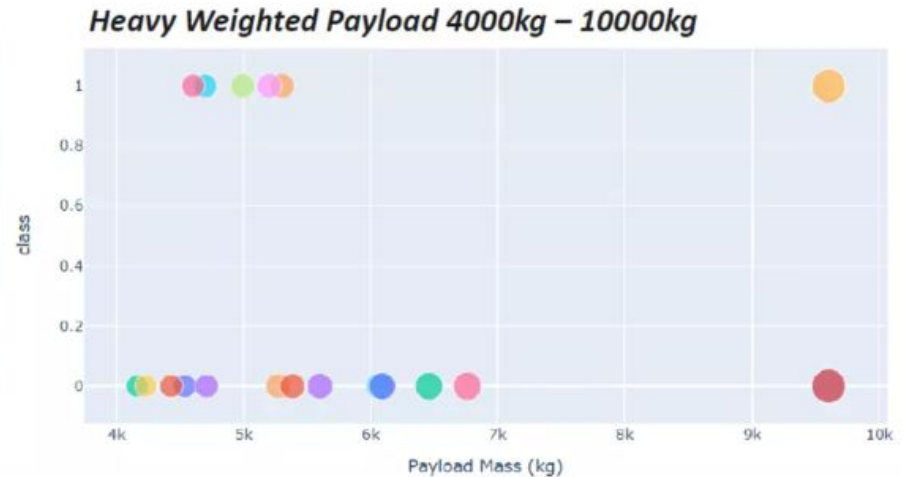
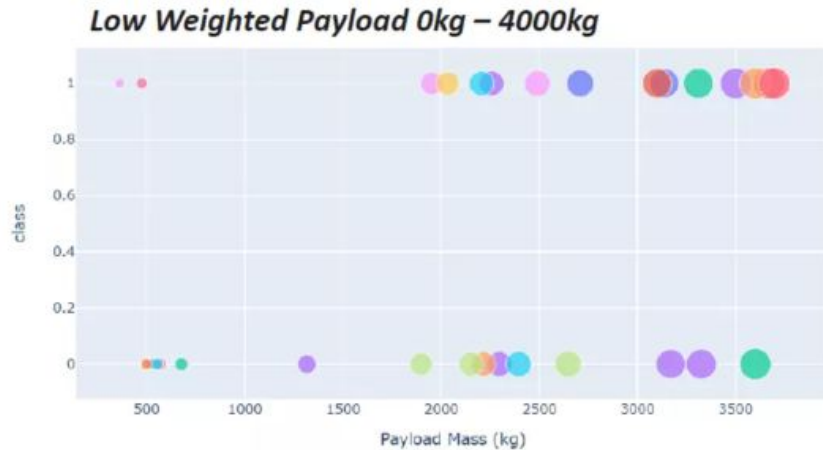
We can see that KSC LC-39A had the most successful launches from all the sites

Success rate by site



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate.

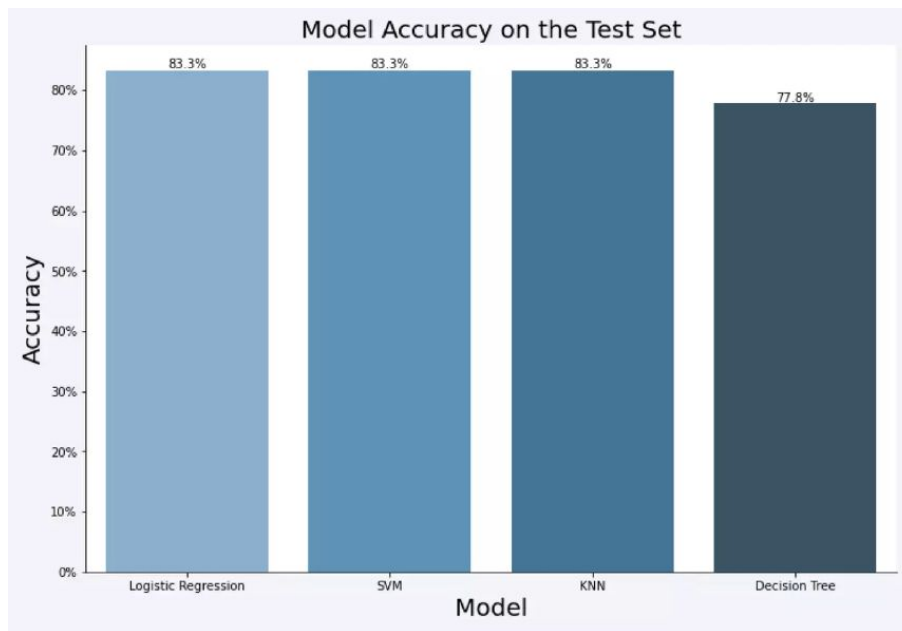
Payload vs Launch outcome



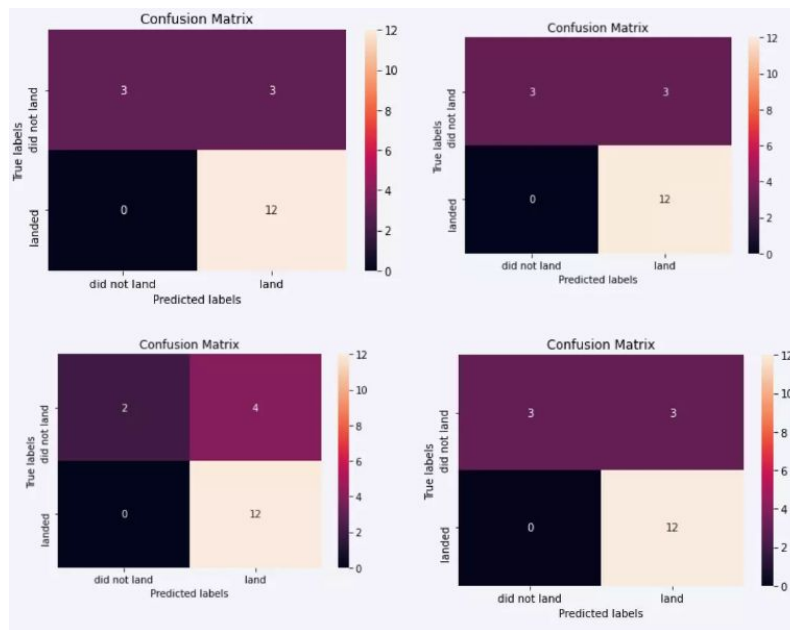
We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix





Conclusions

- The SVM, KNN, and Logistic Regression models are the best in terms of prediction accuracy for this dataset
- Low weighted payloads perform better than the heavier payloads
- The success rates for SpaceX launches has been constantly grow through the years, they will eventually perform perfect launches.
- KSC LC 39A had the most successful launches from all the sites
- Orbit GEO, HEO, SSO, ES L1 had more successful launches.



Thank You!